

Blanca Martin-Burgos

Computational Neuroscientist — Biosignal Analysis & Neural Data Science

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PhD candidate with 6+ years of research experience spanning computational and experimental neuroscience, from single-neuron electrophysiology to network-level and circadian dynamics. My work focuses on extracting interpretable, quantitative structure from noisy physiological time-series data and building reproducible analysis pipelines that link multi-scale signals to system state and function. I manage overlapping wet-lab and computational projects end to end — from data collection through publication — and collaborate readily across disciplines and labs.

POSITIONS AND EMPLOYMENT

PhD Candidate Researcher — UC San Diego, La Jolla, CA

2020 – Present

Advisors: Bradley Voytek, Alysson Muotri

- Parameterized and modeled action potential waveform variability within single neurons, linking intracellular spike shape dynamics to extracellular oscillatory activity and brain state
- Investigated neuronal timescales as an organizing principle of brain function, quantifying how circuits integrate information across systems and developmental stages
- Improved electrophysiology and analysis methods for human cortical organoids to characterize functional neural networks in health and disease, including how stimulation and psychedelics modulate plasticity
- Managed multiple wet-lab and computational research projects in parallel, resulting in peer-reviewed publications, cross-lab collaborations, and conference presentations

Undergraduate Student Researcher — Smith College, Northampton, MA

2017 – 2020

Advisor: Mary Harrington

- Developed a technique to measure circadian bioluminescence from freely behaving animals, and computational tools for analyzing longitudinal circadian time-series data

Undergraduate Student Researcher — Massachusetts Institute of Technology, Cambridge, MA

2019

Advisor: Emery Brown

- Analyzed multivariate neural spiking data with simultaneous events using point-process theory and statistical modeling

EDUCATION

PhD in Neurosciences — UC San Diego, La Jolla, CA

2020 – Present

Computational Neuroscience Specialization · Expected defense: Fall 2026

B.A., Neuroscience & Computer Science — Smith College, Northampton, MA

2015 – 2019

Graduated with Highest Honors

FELLOWSHIPS, GRANTS & AWARDS

Major Competitive Fellowships

CIRM Stem Cell Biology Predoctoral Training Fellowship (2020–2024) — ~\$36,000/yr salary support, full tuition remission, ~\$6,000/yr research & travel supplies

Washington University in St. Louis, Olin Fellowship for Women in Graduate Study (2020, awarded — declined) — ~\$36,720/yr stipend, full tuition remission, travel support

Institutional & Research Fellowships

McKinley Honors Research Fellowship (2019–2020) — \$2,770/yr stipend

Smith College Summer Undergraduate Research Fellowship (2018, 2020) — \$4,800 stipend, awarded twice

Nancy Kershaw Tomlinson Memorial Research Fund (2019)

Honors & Conference Awards

Neuroscience Department Highest Honors (2020) — Thesis: Computational Analysis of in vivo Circadian Bioluminescence in Freely Moving Mice

AEMES Conference Funding Award (2019)

Schultz Foundation Conference Travel Award (2018, 2019)

PUBLICATIONS

Martin-Burgos, B., McPherson, T. S., Hammonds, R., Gao, R., Muotri, A. R., & Voytek, B. (2024). Development of neuronal timescales in human cortical organoids and rat hippocampus dissociated cultures. *Journal of Neurophysiology*, 132(3), 757–764.

LaMontagne, E., Gonzalez, G., Vatsyayan, R., **Martin-Burgos, B.**, Puppo, F., Biagi, D., et al., & Engler, A. J. (2025). Graphene-polymer nanofibers enable optically induced electrical maturation in stem cell-derived cardiomyocytes and brain organoids. *Biomaterials*, 323.

Dixon, T., **Martin-Burgos, B.**, Puppo, F., Chu, T., & Muotri, A. (2023). Modeling ketamine effects on in vitro 3D human neuronal cultures. *Biological Psychiatry*, 93(9), S117–S118.

Martin-Burgos, B., Wang, W., William, I., Tir, S., Mohammad, I., Javed, R., et al., & Leise, T. L. (2022). Methods for detecting PER2:LUCIFERASE bioluminescence rhythms in freely moving mice. *Journal of Biological Rhythms*, 37(1), 78–93.

Smith, C. B., van der Vinne, V., McCartney, E., Stowie, A. C., Leise, T. L., **Martin-Burgos, B.**, et al., & Weaver, D. R. (2022). Cell-type-specific circadian bioluminescence rhythms in Dbp reporter mice. *Journal of Biological Rhythms*, 37(1), 53–77.

van der Vinne, V., **Martin-Burgos, B.**, Harrington, M. E., & Weaver, D. R. (2020). Deconstructing circadian disruption: Assessing the contribution of reduced peripheral oscillator amplitude on obesity and glucose intolerance in mice. *Journal of Pineal Research*, 69(1), e12654.

Cenek, L., Klindziuk, L., Lopez, C., McCartney, E., **Martin-Burgos, B.**, Tir, S., et al., & Leise, T. L. (2020). CIRCADA: Shiny apps for exploration of experimental and synthetic circadian time series with an educational emphasis. *Journal of Biological Rhythms*, 35(2), 214–222.

Martin-Burgos, B., Juavinett, A. L., Riviere, P. D., Hammonds, R., & Voytek, B. (in prep). Action potentials are meaningfully non-binary.

Martin-Burgos, B., Riviere, P. D., Hammonds, R., Juavinett, A. L., & Voytek, B. (in prep). Action potential waveforms can be state dependent.

TEACHING EXPERIENCE & MENTORSHIP

Teaching Assistant, Introduction to Python for Biologists (BILD 62) — UC San Diego

- Led weekly discussion sections (4×/week, 100+ students) on Python programming and data analysis for life sciences; prepared assignments, graded coursework, and supported students through regular office hours

Research Mentorship — UC San Diego and Smith College

- Mentored undergraduate and master's students in neural data analysis, electrophysiology methods, and research best practices
- Mentee received honorable mention for the CRA (Computing Research Association) Undergraduate Research Award

Private Tutoring

- Tutored middle and high school students in mathematics, science, and Spanish

CONFERENCE PRESENTATIONS

Martin-Burgos, B., et al. (2025, October). Beyond all-or-nothing: Intracellular action potential waveform varies systematically with causal stimulation and extracellular oscillatory state. *Society for Neuroscience (SfN)*, San Diego, CA.

Martin-Burgos, B., et al. (2024, October). Not all-or-nothing: Intracellular action potential waveform varies systematically with causal stimulation and extracellular oscillatory state. *SfN*, Chicago, IL.

Martin-Burgos, B., et al. (2024, June). Not all-or-nothing: Intracellular action potential waveform varies systematically with extracellular gamma oscillatory state. *FENS*, Vienna, Austria.

Martin-Burgos, B., et al. (2024, April). Not all-or-nothing: Intracellular action potential waveform varies systematically with extracellular gamma oscillatory state. *Cognitive Neuroscience Society (CNS)*, Toronto, Canada.

Martin-Burgos, B., Puppo, F., Ganapathy, S., et al. (2023, June). Parameterization of neuronal action potential waveform in high-density MEA and patch clamp recordings of control and autistic-patient derived human cortical organoids. *ISSCR*, Boston, MA.

Martin-Burgos, B., McPherson, T. S., et al. (2022, November). Neuronal timescales across the lifespan: Using human cortical organoids to study neurodevelopment and aging. *SfN*, San Diego, CA.

Martin-Burgos, B., McPherson, T. S., et al. (2022, March). Development of neuronal timescales in human cortical organoids and rat hippocampus dissociated cultures. *CNS*, San Francisco, CA.

Martin-Burgos, B., McCartney, E., Weaver, D. R., & Harrington, M. E. (2020, June). Effects of food restriction on circadian rhythms of gene expression recorded in vivo using DBP reporter mice. *SRBR*, Virtual.

Martin-Burgos, B., et al. (2019, October). Measuring circadian bioluminescence from mice lacking a functional central clock. *SfN*, Chicago, IL.

Wang, W., William, I., Smith, S., **Martin-Burgos, B.**, et al. (2018, May). Measuring circadian bioluminescence from freely behaving mice. *SRBR*, Jacksonville, FL.

WORKSHOPS

Martin-Burgos, B., Juavinett, A. L., & Tan, T. (Speakers); Harrington, M. (Organizer/Moderator) (2026, November). Building AI-Aware Assignments for Neuroscience Education. Professional Development Workshop, *Society for Neuroscience (SfN)*, San Diego, CA.

Workshop Assistant, “A hands-on technical workshop for cognitive neuroscientists” (2024, April). *Cognitive Neuroscience Society (CNS)*, Toronto, Canada.

SKILLS

Signal Processing & Time-Series Analysis

Spectral analysis, filtering, time–frequency methods, wavelets, point-process modeling, multivariate time-series analysis, feature extraction from noisy physiological signals

Computational & Statistical Methods

Machine learning, statistical modeling, experimental design, quantitative hypothesis testing, model validation

Programming & Data Analysis

Python (NumPy, SciPy, pandas), MATLAB, R; reproducible analysis pipelines; large-scale data analysis

Experimental & Biosignal Methods

Electrophysiology (patch clamp, MEA), neural signal acquisition and analysis, tissue culture, rodent surgeries and behavioral assays

Tools & Workflow

Git and collaborative development; project and data management for multi-disciplinary research

Languages

Spanish (Native), English (Bilingual, C2), French (Advanced, C1), German (Beginner, B1)

U.S. work authorization: F-1 OPT (no sponsorship required at this time)